Architectural Blueprints for High-Stakes Public Infrastructure: Engineering Autonomous Multi-Agent State Machines for Systemic American Crises

Admissions evaluation committees at institutions such as the Massachusetts Institute of Technology, Stanford University, and Carnegie Mellon University, alongside national science juries for the Regeneron Science Talent Search and the International Science and Engineering Fair, evaluate candidate computing portfolios through the lens of systems engineering, algorithmic determinism, and civic consequence. Across elite academic panels, projects characterized by superficial generative wrappers—such as conversational interfaces over static documents or trivial prompt-forwarding scripts—are systematically dismissed. Genuine distinction requires the design of asynchronous, distributed architectures that enforce formal verification routines and interface with United States public systems and open-data standards. The five system blueprints presented below address severe, day-to-day crises across the American domestic landscape: predatory healthcare billing, municipal drinking water toxicity, tenant exploitation through constructive eviction, special education civil rights compliance, and critical municipal emergency bottlenecks. Each system is architected within a bounded, production-grade technology profile: modern React.js and Tailwind CSS for client-side state visualization, asynchronous Python 3.11+ and FastAPI for REST ingestion, PostgreSQL with langgraph-checkpoint-postgres (AsyncPostgresSaver) for durable state persistence, and LangGraph for cyclic agentic orchestration using the Model Context Protocol (MCP)[1].

Project 1: EquiClaim — Autonomous Cross-Examination Engine for Insurance Denial Adjudication and Hospital Price Transparency Compliance

1. The Human & Systemic Crisis (The "Admissions Hook")

Medical debt is the leading cause of personal bankruptcy in the United States, directly affecting more than one hundred million individuals. Working-class families routinely receive balance bills ranging from $5,000 to over $100,000 following emergency room visits or inpatient surgical procedures. These unexpected financial obligations are driven by automated claim adjudication engines deployed by commercial health insurers that systematically reject coverage using standardized Claim Adjustment Reason Codes (CARC) and Remittance Advice Remark Codes (RARC). Concurrently, hospital billing departments frequently disregard federal mandates under the Centers for Medicare & Medicaid Services (CMS) Hospital Price Transparency regulation (45 CFR § 180) and the No Surprises Act (45 CFR § 149), obscuring discounted cash prices and billing patients well above the statutory Qualifying Payment Amount (QPA)[5].

The status quo persists because dispute resolution operates as an asymmetric war of attrition. An ordinary citizen cannot parse multi-gigabyte machine-readable files, decipher five-digit

Current Procedural Terminology (CPT) modifiers, or navigate the procedural deadlines of the federal Independent Dispute Resolution (IDR) process. As a result, patients often deplete life savings or yield to aggressive collection agencies for debts that are legally invalid or grossly inflated under federal statute.

2. Verified US Open Data Anchors

The system directly ingests and cross-references multiple federal pricing disclosures, regulatory tables, and standardized billing registries.

Data Source Anchor	Public Endpoint / Source URL	Format and Schema Profile	Cadence and Volume
CMS Hospital Price Transparency Machine-Readable Files	Mandatory hospital web postings under 45 CFR § 180; validated via CMS Online Validator Tool[5]	CMS MRF v2 Schema; CSV and JSON; nested pricing hierarchies[5]	Updated annually (monthly revisions); over 6,000 hospital endpoints; multi-gigabyte files[5]
CMS Physician Fee Schedule (PFS)	CMS.gov / Data.gov REST endpoints and relative value units (RVU) flat files	Tabular relational data; CPT/HCPCS codes with geographic practice cost indices (GPCI)	Updated annually with quarterly statutory adjustments; >10,000 active service codes
X12 Standardized Denial Registries	Washington Publishing Company (WPC) official registries	XML and JSON definitions of CARC and RARC claim adjustment reason codes	Triannual formal updates; hundreds of standardized reason and remark codes
CMS National Correct Coding Initiative (NCCI)	Data.gov and CMS.gov NCCI Policy Manual and edit tables	Relational CSV tables; Procedure-to-Procedure (PTP) edits and Medically Unlikely Edits (MUE)	Quarterly releases; >500,000 code-pair combinations determining unbundled billing rules

The structural complexity of this raw data prevents lay interpretation. Hospital machine-readable files conforming to the CMS MRF v2 specification feature deeply nested relational matrices where a single procedure code branches into gross charges, discounted

cash rates, and hundreds of payer-specific negotiated charges[5]. Furthermore, billing departments exploit modifier nuances—such as appending Modifier -25 or -59—to bypass National Correct Coding Initiative unbundling edits. Determining whether a balance bill violates the No Surprises Act requires calculating the median in-network contracted rate for the geographic area (the statutory QPA) across historical claims, an operation that cannot be conducted without specialized data infrastructure.

3. Why a Multi-Agent State Machine is Architecturally Mandatory

A single language model prompt or linear script fails when applied to forensic medical billing due to arithmetic hallucinations, context window overflow across large data files, and an inability to execute multi-step regulatory logic. Resolving a disputed bill requires an asynchronous StateGraph where specialized sub-agents cross-examine the claim from competing analytical perspectives[3].

Pipeline Node Identity	Node Classification	Operational Responsibility	Input Dependencies	Emitted State Channels
Intake & CARC/RARC Forensic Worker	Worker Node	Normalizes scanned EOBs/bills, parses CPT/HCPCS line items, maps CARC/RARC codes	Digitized billing documents	claim_line_items, denial_mechanisms
MRF & Fee Schedule Benchmark Worker	Worker Node	Executes parameterized SQL over local PostgreSQL tables storing hospital MRFs to isolate cash and median rates	claim_line_items, hospital NPI	mrf_cash_rate, qpa_geographic_median
No Surprises Act Compliance Worker	Worker Node	Audits provider network status and facility emergency classifications	claim_line_items, facility metadata	statutory_violations, idr_eligibility

		against 45 CFR § 149 balance-billing rules		
Actuarial Invariant Evaluator Node	Evaluator-Optimizer Loop	Enforces exact integer arithmetic in cents, verifies statutory citations, and checks legal dispute grounds[3]	Worker outputs across all channels	certified_audit_docket, rejection_vector

The Evaluator-Optimizer loop operates as a formal verification boundary[3]. When the specialized workers generate dispute grounds and financial reconciliations, the Evaluator Node verifies that every dollar sum matches the integer sum (in cents) of its constituent line items. It confirms that the derived QPA benchmark corresponds to the median contracted rate for the specific geographic metropolitan statistical area and checks that dispute arguments cite valid paragraphs of 45 CFR § 149.

If an agent misinterprets an NCCI unbundling exception or proposes a settlement below Medicare base reimbursement levels, the Evaluator rejects the graph state, appends targeted correction instructions to the state channel, and routes execution back to the worker nodes for recalculation[3]. Durable graph execution is managed by LangGraph's AsyncPostgresSaver, which records execution snapshots across checkpoints, checkpoint_blobs, and checkpoint_writes tables[1]. This enables multi-day workflows where patients pause the review process to retrieve itemized billing codes from a hospital, resuming the state machine without loss of context[1].

4. Admissions Impact & Faculty Interview Narrative

The candidate frames the project as a direct challenge to the administrative friction used to extract wealth from vulnerable families. The narrative describes a progression from observing a family member immobilized by an inexplicable out-of-network emergency bill to discovering that the data required to invalidate the charge was buried in federal open-data repositories. This narrative demonstrates intellectual independence, civic engagement, and an ability to translate public policy statutes into software systems.

During an interview, a faculty member may examine the decision to use an asynchronous multi-agent graph with relational checkpointing rather than a standard retrieval-augmented generation (RAG) pipeline. The candidate defends the design on the grounds of mathematical determinism and legal precision. Semantic vector search operates on statistical proximity within continuous embedding spaces, which inherently struggles with exact numerical thresholds, procedural modifiers, and statutory fee calculations. Missing a single CPT modifier

or miscalculating a median rate by three cents invalidates a legal dispute.
Unstructured document parsing is therefore isolated to the edge workers, while all legal invariants, NCCI unbundling rules, and financial calculations are executed via parameterized relational SQL in PostgreSQL and verified by a cyclic Evaluator-Optimizer loop[3]. LangGraph's AsyncPostgresSaver guarantees state durability across network interruptions, providing complete computational reproducibility for legal arbitration[1].

5. User Experience (React + FastAPI Integration)

A patient or legal aid caseworker uploads a photograph or digital PDF of an itemized hospital bill or insurance EOB through a modern React interface styled with Tailwind CSS. FastAPI handles file ingestion through asynchronous endpoints, validating document metadata via strict Pydantic v2 schemas before creating a LangGraph thread. The client maintains an active polling or streaming connection managed via custom React Query hooks.
The dashboard provides real-time visibility into the system's reasoning:

- A visual execution trace displays the active state of the LangGraph topology, tracking progress as the engine transitions from OCR tokenization to parallel database queries against the hospital's multi-gigabyte MRF dataset.
- An interactive Forensic Audit Ledger presents an itemized reconciliation comparing the hospital's gross charges against the facility's published discounted cash rate, the regional median in-network rate, and Medicare fee schedules.
- A statutory dispute panel highlights detected legal violations, such as unbundled imaging fees or illegal balance billing for emergency pathology services under 45 CFR § 149.
- A single click generates an audit docket containing a formal notice of dispute addressed to the insurer and hospital compliance officer, cited directly to federal regulations and backed by accessioned MRF data.

6. Feasibility Audit (4-Week Engineering Sprint)

The minimum viable product limits data ingestion to emergency medicine claims (CPT codes 99281 through 99285) for three major hospital systems within a single state. The CMS MRF v2 CSV files for these facilities are ingested, normalized, and indexed into relational PostgreSQL tables[5]. The three-worker LangGraph pipeline is compiled using AsyncPostgresSaver to evaluate balance billing violations and generate dispute filings[1].
The primary failure mode stems from non-standard CSV formatting, encoding errors, and broken schema links within hospital MRF disclosures[5]. The ingestion layer defensively handles this using resilient Python stream-parsers that sanitize headers, reject malformed rows, and isolate schema variances into a quarantine database table. If an MRF lookup fails or times out, the system triggers a circuit breaker that falls back to statutory CMS Physician Fee Schedule regional baselines, notifying the evaluator node of the fallback state.

Project 2: AquaSentry — Multi-Agent Hydro-Toxicology Sentinel for Municipal Lead Service Lines and PFAS Exposure Remediation

1. The Human & Systemic Crisis (The Admissions Hook)

Access to safe drinking water is a fundamental human necessity, yet millions of Americans across thousands of municipalities consume tap water contaminated with neurotoxic lead and

per- and polyfluoroalkyl substances (PFAS). A decade after the crisis in Flint, Michigan, hundreds of thousands of legacy lead service lines remain unmapped and buried beneath low-income neighborhoods. Concurrently, synthetic PFAS compounds—linked to renal carcinomas, thyroid disruption, and immunological deficits—accumulate in municipal aquifers without public awareness.

The status quo fails because municipal water utilities publish annual Consumer Confidence Reports (CCRs) that obscure localized danger. These reports rely on regional aggregated averages, masking severe contamination spikes that occur at specific school drinking fountains, residential branch lines, or environmental justice corridors.

Low-income households cannot afford private mass-spectrometry laboratory tests ($400–800 per sample), nor can they decipher the chemical nomenclature and parts-per-trillion (ppt) metrics required to challenge municipal water quality assertions under the Safe Drinking Water Act (SDWA).

2. Verified US Open Data Anchors

The system integrates federal drinking water compliance repositories, live hydrological telemetry, and statutory chemical monitoring inventories.

Data Source Anchor	Public Endpoint / Source URL	Format and Schema Profile	Cadence and Volume
EPA Safe Drinking Water Information System (SDWIS)	EPA Envirofacts REST API (https://data.epa.gov/efservice/)	JSON and XML REST formats; public water system violation schemas	Quarterly compliance releases; >148,000 public water systems cataloged
USGS National Water Information System (NWIS)	USGS Water Quality Portal REST Services (https://www.waterqualitydata.us/)	GeoJSON and WQX schema; physical and geochemical water attributes	Real-time and discrete historical samples; >100 million sensor records nationwide
EPA Unregulated Contaminant Monitoring Rule (UCMR 5)	EPA UCMR 5 Data Releases (https://www.epa.gov/dwucmr/)	Bulk CSV and relational flat files; 29 PFAS congeners and lithium concentrations	Semiannual reporting updates; covers all public systems serving >3,300 people
EPA Lead and Copper Rule Improvements	EPA LCR Compliance Files and Municipal Open	GeoJSON spatial vectors and tabular service line material	Annual updates; millions of mapped public and private pipe material

(LCRI) Inventories	GIS Portals	declarations	segments

The complexity of this data prevents manual interpretation by non-specialists. SDWIS datasets organize water systems hierarchically under Public Water System Identification (PWSID) numbers, separating sampling points, analytical methods, and violation tiers into disparate relational tables. Furthermore, evaluating toxicological risk requires reconciling discrete sample points across varying measurement units:

$$\text{Parts per Billion (ppb)} \equiv \mu\text{g/L}, \quad \text{Parts per Trillion (ppt)} \equiv \text{ng/L}$$

Correlating these values with dynamic geochemical factors, such as pH and alkalinity fluctuations that trigger lead leaching, requires automated spatial and toxicological processing.

3. Why a Multi-Agent State Machine is Architecturally Mandatory

Evaluating drinking water safety requires unifying continuous hydrological science, spatial topology, and changing federal chemical regulations. A single generative model or rigid script cannot accurately evaluate these intersections:

Pipeline Node Identity	Node Classification	Operational Responsibility	Input Dependencies	Emitted State Channels
SDWIS Regulatory Compliance Worker	Worker Node	Queries EPA SDWIS for system-specific Maximum Contaminant Level (MCL) exceedances	System PWSID, geographic coordinates	active_violations, historical_infractions
USGS Geochemical & Hydrological Worker	Worker Node	Retrieves upstream surface/groundwater telemetry, evaluating corrosive leaching parameters	Hydrological Unit Codes (HUC-8/12)	geochemical_profiles, corrosivity_indices

Demographic & Spatial Exposure Worker	Worker Node	Intersects municipal GIS pipe materials with Census ACS data and school boundary shapefiles	Municipal GIS layers, census tracts	exposure_vulnerability_score, impacted_schools
Toxicological Invariant Evaluator Node	Evaluator-Optimizer Loop	Enforces strict unit conversions, verifies statutory MCL/MCLG thresholds, and audits spatial risk assertions[3]	Worker outputs across all channels	certified_water_docket, re-evaluation_directive

The Evaluator-Optimizer loop prevents hallucinations regarding chemical safety[3]. When workers extract and evaluate chemical telemetry, the Evaluator validates unit consistency across disparate reporting standards and compares reported PFAS concentrations against the EPA's enforceable National Primary Drinking Water Regulation Maximum Contaminant Levels (e.g., 4.0 ppt for PFOA and PFOS).

If a worker mistakenly classifies a 15 ppb lead measurement as safe because it fell below an aggregated regional average, the Evaluator catches the error, marks the finding as an explicit violation of the Lead and Copper Rule action level (15 ppb), and routes the task back to the worker to adjust the risk posture[3]. State persistence is maintained using AsyncPostgresSaver, saving intermediate graph steps across checkpoints and checkpoint_writes[1]. This allows long-running geographic queries over USGS datasets to proceed without holding open active HTTP connections, ensuring system resilience even under poor network conditions[1].

4. Admissions Impact & Faculty Interview Narrative

The candidate presents the system as a computational response to structural environmental neglect, detailing how low-income neighborhoods often drink water with elevated corrosivity and lead levels while affluent suburbs benefit from advanced chemical filtration. The narrative highlights how systems engineering can convert obscure environmental data into actionable civic legal action, illustrating ethical commitment and advanced technical capability.

During a technical review, a faculty evaluator may inquire about how the architecture handles spatial interpolation of water quality between monitored USGS sampling stations without hallucinating contamination plumes. The candidate explains that the language model is never permitted to perform spatial interpolation or predict chemical dispersion. Instead, deterministic

GIS operations within PostgreSQL (PostGIS) execute inverse-distance weighting and buffer intersections over validated USGS sampling geometries.
The LangGraph agents only interpret the deterministic output of these spatial queries against the statutory frameworks of the Safe Drinking Water Act[3]. State continuity is managed asynchronously using AsyncPostgresSaver, isolating tool latency and guaranteeing that toxicological assertions are grounded in verified federal sensor records[1].

5. User Experience (React + FastAPI Integration)

A resident or public school administrator inputs a street address or school facility name into a React frontend styled with Tailwind CSS. FastAPI validates the query through Pydantic v2 schemas and assigns the request to a unique LangGraph thread identifier.
The resulting dashboard provides clear, actionable environmental insights:

- An interactive Mapbox/Leaflet spatial interface displays local water distribution networks, highlighting known or suspected lead service lines in red, copper lines in green, and unmapped lines in amber.
- A live Agent Reasoning Feed illustrates the multi-agent investigation in real time, showing the system querying SDWIS APIs, retrieving USGS raw water samples, and checking UCMR 5 PFAS levels.
- A Toxicity vs. Regulation Breakdown contrasts the utility's reported chemical concentrations against both enforceable EPA Maximum Contaminant Levels (MCLs) and health-based Maximum Contaminant Level Goals (MCLGs).
- An automated Citizen Legal Notification Generator drafts formal petitions under 40 CFR § 141.85, empowering residents to demand immediate water testing and service line replacement from their municipal water board.

6. Feasibility Audit (4-Week Engineering Sprint)

The sprint MVP scopes data ingestion to a single vulnerable urban watershed (such as Newark, New Jersey or Flint, Michigan). The system ingests historical SDWIS violation tables and public lead service line inventory GeoJSON datasets into a PostGIS-enabled PostgreSQL instance. The three specialized workers and the toxicological validation loop are deployed to generate localized lead exposure reports.
The primary failure mode involves SDWIS API endpoints experiencing high latency or returning empty records due to state-level reporting delays. The architecture handles this through an asynchronous fallback pattern: if live EPA REST calls exceed a 2,500ms timeout threshold, the system reads from a locally replicated PostgreSQL mirror of the federal database, flagging the report with the exact date of the local snapshot to preserve audit integrity.

Project 3: HabitaLex — Municipal Code Enforcement Arbiter and Constructive Eviction Forensic Defense Pipeline

1. The Human & Systemic Crisis (The Admissions Hook)

Across major American metropolitan centers, housing instability is driven not only by formal court evictions, but by covert constructive evictions. Predatory landlords deliberately neglect rent-stabilized properties—shutting off heating during freezing winter months, allowing black mold to spread, and ignoring structural roof leaks—to force low-income tenants out so units can be renovated and re-leased at market rates.

When vulnerable families attempt to defend themselves in municipal housing courts, they enter an unbalanced system: landlords are represented by specialized legal counsel, while over 85% of tenants navigate complex eviction proceedings without legal representation.
Although municipalities maintain public records of building code violations and 311 emergency complaints, this civic data is locked in siloed open-data portals. An unrepresented tenant cannot easily cross-reference hundreds of building code infractions, trace opaque limited liability company (LLC) ownership networks, or link municipal inspection tickets to the implied warranty of habitability under state property laws. As a result, tenants are routinely evicted for non-payment of rent, even when their living conditions violate basic housing laws.

2. Verified US Open Data Anchors

The system links municipal code enforcement streams, public complaint repositories, and corporate property registration databases.

Data Source Anchor	Public Endpoint / Source URL	Format and Schema Profile	Cadence and Volume
NYC OpenData / HPD Building Violations	NYC OpenData Socrata SODA API (https://data.cityofnewyork.us/resource/wvxf-dwi5.json)	JSON REST and tabular CSV; Class A, B, and C violation schema	Daily automated updates; >5 million cumulative building violation records
Municipal 311 Tenant Complaints	Socrata Open Data Networks (NYC, Chicago, DataSF 311 streams)	GeoJSON and REST JSON; emergency heating and infrastructural complaint logs	Real-time / daily updates; tens of thousands of service requests logged monthly
HUD Real Estate Assessment Center (REAC)	HUD Open Data Portal (https://data.hud.gov/)	Tabular relational files; physical inspection scores and health deficiency indices	Annual releases; covers all federally subsidized and public housing properties
County Recorder of Deeds / Property Assessment	County Assessor open databases (e.g., NYC ACRIS, Cook County Portal)	Relational property deed transfers, mortgage recordings, and corporate filings	Continuous daily filings; millions of transaction documents resolving parcel ownership

The primary obstacle in this data space is intentional corporate obscurity. Real estate investors purchase residential parcels under isolated, single-asset LLCs (e.g., "123 North 7th Street LLC"), masking beneficial ownership and systemic patterns of neglect across portfolios. Furthermore, municipal violation classifications (such as Class A non-hazardous, Class B hazardous, and Class C immediately hazardous under New York law) require precise mapping to statutory habitability defenses to establish valid counterclaims in court.

3. Why a Multi-Agent State Machine is Architecturally Mandatory

Building a constructive eviction defense requires reconciling unstructured tenant testimony, civic inspection databases, and property deed records. A simple single-prompt model fails due to corporate entity resolution errors, poor date arithmetic, and hallucinations of municipal housing codes:

Pipeline Node Identity	Node Classification	Operational Responsibility	Input Dependencies	Emitted State Channels
Municipal Code & Inspection Worker	Worker Node	Interfaces via FastMCP to query municipal APIs, isolating active hazardous building violations	Parcel identifier (BBL/APN), address	active_violations, inspection_history
LLC De-Anonymization Worker	Worker Node	Traverses deed records and corporate filings to identify the beneficial owner and portfolio-wide infractions	Owner name, corporate entity filings	beneficial_owner, portfolio_violation_rate
Habitability Defense Worker	Worker Node	Evaluates violation severity and duration against state property law to	active_violations, lease payment records	abatement_calculations, defense_pleadings

		calculate legal rent abatements		
Judicial Evidentiary Evaluator Node	Evaluator-Optimizer Loop	Audits claims against official inspection records and verifies rent-withholding escrow math[3]	Worker outputs across all channels	verified_legal_defense_docket, audit_retry

The Evaluator-Optimizer loop functions as an evidentiary quality-control gate[3]. In housing court proceedings, filing an unverified counterclaim can result in an immediate summary judgment against the tenant. The Evaluator Node checks that every cited building code violation corresponds to an active, documented municipal notice with an official inspection ID. It recalculates the proposed rent abatement according to strict statutory percentage formulas:

$$\text{Allowable Abatement} = \sum_{i} \left(\text{Monthly Rent} \times \text{Severity Weight}_i \times \frac{\text{Days Out of S}}{\text{Days in M}} \right.$$

If an agent miscalculates the elapsed days of a heating outage or cites an unrelated zoning ordinance instead of a housing maintenance code, the Evaluator rejects the output and triggers a corrective refinement step[3].

State persistence is maintained through LangGraph's AsyncPostgresSaver, saving intermediate execution states within PostgreSQL[1]. This architecture ensures that when caseworkers gather evidence over several weeks, the agent state remains secure, indexed, and fully auditable[1].

4. Admissions Impact & Faculty Interview Narrative

The candidate frames the project around bridging the legal representation gap through computational systems. The narrative describes attending a local housing court and watching families lose their homes simply because they lacked the technical tools to access and present municipal public data. This narrative demonstrates social maturity, civic responsibility, and an understanding of how software can safeguard fundamental civil rights.

During technical evaluations, a faculty evaluator may ask how the system resolves the identity of corporate landlords across fragmented LLCs without generating false-positive links between unrelated property owners. The candidate explains that probabilistic string matching over corporate names is avoided. Instead, the architecture executes deterministic graph traversals in PostgreSQL to join business entities sharing identical registered agent addresses, corporate officer names, and business filing numbers from state corporate registries.

The LangGraph agent only receives clustered and verified corporate nodes, using its reasoning capacity to synthesize patterns of portfolio-wide neglect rather than guessing ownership links. LangGraph's checkpointer guarantees that all entity-linkage steps are fully auditable and reproducible for court presentation[1].

5. User Experience (React + FastAPI Integration)

A tenant or legal-aid caseworker accesses the system via a responsive React web interface styled with Tailwind CSS. The tenant inputs their building address, monthly rent amount, and a brief description of unaddressed conditions (such as missing heat or ceiling leaks).
FastAPI orchestrates the background analysis asynchronously, streaming agent updates to the client:

- An interactive Building Violation Timeline displays past and present municipal citations, color-coded by legal severity (Class A, B, or immediately hazardous Class C).
- An LLC Landlord Network Map reveals the landlord's broader property portfolio, illustrating patterns of systemic code violations across other properties held by the same beneficial owner.
- A Habitability Abatement Calculator displays estimated rent offsets based on the exact duration and severity of unaddressed infractions.
- An exportable Court-Ready Defense Packet produces an official Housing Part (HP) action petition, complete with municipal citation accession numbers and statutory references, ready for submission to a housing court judge.

6. Feasibility Audit (4-Week Engineering Sprint)

The sprint MVP focuses on New York City, integrating the NYC OpenData Socrata APIs for HPD violations and 311 service requests. The multi-agent StateGraph is deployed to ingest an address, identify unresolved Class C hazardous violations, compute rent abatement percentages, and export a formal housing court answer document.
The primary failure mode involves municipal Socrata open-data endpoints enforcing strict rate limits (HTTP 429 Too Many Requests) during bulk querying. The system handles this by using an asynchronous token-bucket rate limiter alongside a local PostgreSQL cache with an indexed 7-day Time-To-Live (TTL). This ensures repeat queries for the same residential parcel resolve locally without failing.

Project 4: LexAdvocate — Algorithmic IDEA Compliance Auditor and Disproportionality Forensic Engine for Public School Special Education

1. The Human & Systemic Crisis (The Admissions Hook)

Under the federal Individuals with Disabilities Education Act (IDEA, 20 U.S.C. § 1400), over seven million American public school children with physical, cognitive, and emotional disabilities have a legal right to a Free Appropriate Public Education (FAPE) via an Individualized Education Program (IEP)[10]. In practice, school districts face persistent staffing shortages, budget constraints, and administrative backlogs, resulting in systemic non-compliance.
Children diagnosed with autism, dyslexia, and ADHD often wait months past statutory evaluation deadlines to receive accommodations. Many are denied critical related

services—such as speech-language pathology, occupational therapy, and dedicated paraprofessionals—without parental knowledge.

Low-income and minority parents face significant structural barriers: they must navigate dense psychological evaluations filled with psychometric statistics (z-scores, standard scores, and percentiles), while school district teams use bureaucratic jargon to deflect resource allocations. When disputes arise, families cannot afford private educational advocates, who charge $150 to $350 per hour.

Compounding this crisis, Department of Education civil rights data reveals that students with disabilities—particularly students of color—face disproportionate disciplinary removals, illegal physical restraint, and improper disciplinary suspensions[10].

2. Verified US Open Data Anchors

The system integrates federal civil rights surveys, special education performance plans, and local school district report cards.

Data Source Anchor	Public Endpoint / Source URL	Format and Schema Profile	Cadence and Volume
US Department of Education Civil Rights Data Collection (CRDC)	Official CRDC Data Portal (https://civilrightsdata.ed.gov/)[10]	Tabular CSV and fixed-width files; disaggregated school-level indicators[10]	Biennial mandatory survey; >17,000 school districts and >97,000 public schools[10]
OSEP State Performance Plan / Annual Performance Reports	ED.gov OSEP SPP/APR Public Database	Relational database extracts; State-level IDEA Indicator 1 through 17 metrics	Annual statutory reporting; covers all 50 state educational agencies
NCES Common Core of Data (CCD)	National Center for Education Statistics API	JSON and CSV REST endpoints; school demographic and fiscal directory data	Annual updates; directory of all operational public elementary and secondary schools
State Educational Agency (SEA) Due Process Rulings	State Department of Education Due Process Hearing Registries	Unstructured legal PDF decisions and administrative dispute logs	Continuous ongoing administrative filings; thousands of state hearing

			transcripts

CRDC datasets contain thousands of variables subject to complex cross-tabulation and privacy suppression rules[10]. Calculating systemic disproportionality—such as whether a school district suspends students with disabilities at rates higher than state baselines—requires applying risk ratios and composition metrics that are difficult for individual parents and lay advocates to calculate without programmatic tooling.

3. Why a Multi-Agent State Machine is Architecturally Mandatory

Evaluating special education compliance requires parsing psychometric metrics, cross-referencing federal legal standards, and comparing local district performance against national civil rights benchmarks. A single LLM prompt fails due to hallucinations of federal regulations, mathematical errors in standard deviation calculations, and an inability to track multi-month procedural deadlines:

Pipeline Node Identity	Node Classification	Operational Responsibility	Input Dependencies	Emitted State Channels
IEP & Psychometric Ingestion Worker	Worker Node	Extracts goals, mandated service minutes, and standardized psychometric scores	Unstructured IEP documents, evaluations	service_commitments, psychometric_profiles
IDEA Statutory Compliance Worker	Worker Node	Audits accommodations and evaluation timelines against 34 CFR Part 300 mandates	service_commitments, referral dates	procedural_violations, fape_deficits
CRDC District Equity Worker	Worker Node	Queries local LEA metrics in the CRDC database via FastMCP, benchmarking equity	School NCES ID, district code	equity_benchmarks, disproportionality_ratios

		indicators[10]		
FAPE Invariant Evaluator Node	Evaluator-Optimizer Loop	Enforces timeline math, psychometric score validation, and OSEP disproportionality schemas[3]	Worker outputs across all channels	certified_iep_audit_docket, retry_flag

The Evaluator-Optimizer loop ensures legal and technical precision[3]. In special education administrative due process hearings, inaccurate claims damage parental credibility. The Evaluator Node verifies that every flagged timeline breach accurately calculates elapsed days:

$$\text{Elapsed Assessment Window} = \text{Evaluation Date} - \text{Parental Conse}$$

It confirms that service shortages reference specific federal regulations (such as 34 CFR § 300.320 for mandatory statements of special education services) and checks that CRDC risk-ratio calculations strictly adhere to the OSEP Significant Disproportionality calculation methodology[10].

If an agent misinterprets a percentile rank or references an obsolete state standard, the Evaluator resets the state and triggers targeted recalculations[3]. AsyncPostgresSaver manages thread persistence in PostgreSQL, storing graph steps within checkpoints and checkpoint_blobs[1]. This enables multi-stage analysis where an advocate can upload historical evaluations, incorporate teacher feedback over several sessions, and maintain a verifiable audit trail without data loss[1].

4. Admissions Impact & Faculty Interview Narrative

The candidate frames the project as an initiative to democratize special education advocacy through systems software. The narrative describes working with neurodivergent students or witnessing a family member struggle to secure legally mandated accommodations against bureaucratic resistance. The narrative demonstrates intellectual maturity, public policy awareness, and an understanding of how automated audit systems can protect vulnerable children.

During an interview, a faculty evaluator may examine the ethical and legal boundaries surrounding the Unauthorized Practice of Law (UPL) when advising parents on IEP disputes. The candidate explains that the system is architected as an algorithmic forensic auditor, not an autonomous legal counsel. It operates deterministically: it extracts documented service commitments, runs mathematical timeline verifications, and benchmarks district metrics against the Department of Education's public Civil Rights Data Collection[10].

The Evaluator-Optimizer loop ensures that every generated output is framed strictly as an empirical factual audit docket, complete with exact statutory citations to 34 CFR Part 300. This provides parents with objective, verified data for formal proceedings while avoiding speculative legal advice.

5. User Experience (React + FastAPI Integration)

A parent or educational advocate accesses the system through an accessible React interface built with Tailwind CSS. The user uploads a digital copy of an IEP or multi-disciplinary evaluation document.

FastAPI processes the file through asynchronous background tasks, initiating a LangGraph audit session.

The resulting dashboard provides clear, actionable special education insights:

- An IEP Service Matrix contrasts mandated instructional hours and therapy allocations against delivered service logs, highlighting unfulfilled commitments.
- A Procedural Timeline Tracker visualizes the evaluation sequence, flagging any violations of the federal 60-day consent-to-evaluation mandate.
- A CRDC Disproportionality Panel illustrates district-wide equity metrics, displaying how the local school compares to state and national averages regarding disciplinary exclusions, law enforcement referrals, and restraint/seclusion rates for students with disabilities[10].
- An automated IDEA Compliance Docket generates a formal Prior Written Notice (PWN) demand letter and administrative due process complaint outline, cited directly to federal statutes and local district performance data.

6. Feasibility Audit (4-Week Engineering Sprint)

The sprint MVP scopes evaluation to IEP documents against federal 34 CFR Part 300 requirements, using historical CRDC data for a single state (such as Ohio or California)[10]. District-level special education enrollments, discipline rates, and related service staffing levels are ingested into PostgreSQL[10]. The three-worker LangGraph pipeline is deployed to audit procedural timelines and calculate basic disproportionality metrics.

The primary failure mode stems from unstructured, scanned, or handwritten IEP PDFs that vary across school districts. The ingestion pipeline handles this with a resilient parsing architecture: document text is extracted and validated through strict Pydantic v2 schemas that enforce structural requirements. Incomplete or ambiguous data fields are marked with confidence flags, prompting the user for review before proceeding through the graph.

Project 5: ResilioNet — Dispatch Optimization and Thermal-Grid Fragility Sentinel for Rural EMS Dead-Zones and Power Grid Bottlenecks

1. The Human & Systemic Crisis (The Admissions Hook)

In rural American communities and economically disadvantaged peri-urban areas, emergency medical services (EMS) face severe systemic bottlenecks. Across large regions of Appalachia, the rural South, and Western tribal reservations, 911 ambulance transit times routinely exceed 20 to 30 minutes—far beyond the critical eight-minute window necessary to prevent brain

death during cardiac arrest or manage severe trauma.
These emergency dead-zones are exacerbated by increasingly severe climate events. Heatwaves, winter storms, and severe weather overload regional electrical transmission corridors, causing distribution feeder trips and substation transformer failures. When regional power grids experience localized blackouts, medically vulnerable individuals who depend on electricity for life-sustaining medical equipment (such as home oxygen concentrators, continuous positive airway pressure machines, and ventricular assist devices) are stranded without warning.
The status quo fails because rural EMS dispatch centers, regional electric utilities, and public health authorities operate in technical silos. Dispatch routing engines do not account for real-time grid telemetry, while electric utilities have little visibility into where power-dependent individuals reside. During extreme weather events, this lack of coordination leads to preventable tragedies.

2. Verified US Open Data Anchors

The system fuses regional power grid telemetry, real-time meteorological feeds, federal medical vulnerability databases, and road network datasets.

Data Source Anchor	Public Endpoint / Source URL	Format and Schema Profile	Cadence and Volume
US EIA Hourly Electric Grid Monitor API v2	US EIA Open Data API (https://api.eia.gov/v2/electricity/rto/)[18]	JSON REST; hourly balancing authority demand, generation, and interchange flows[18]	Hourly continuous updates; covers all Regional Transmission Organizations (RTOs)
NOAA NWS Active Alerts API	National Weather Service API (https://api.weather.gov/alerts)	GeoJSON / CAP (Common Alerting Protocol); active meteorological alert polygons	Event-driven real-time updates; continuous severe weather alerts nationwide
HHS emPOWER Public Map & Datasets	US Health and Human Services emPOWER Portal	Tabular CSV and spatial REST; Medicare claims for electricity-dependent life support	Monthly updates; county and zip code aggregates nationwide
US Department of Transportation	OpenStreetMap Overpass API and	GeoJSON vector road networks;	Continuous / quarterly updates;

(USDOT) / OSM	USDOT National Transportation Atlas	road classifications and bridge restrictions	comprehensive US surface transportation networks

The EIA grid telemetry corpus reflects dynamic physical interactions across balancing authorities, where regional transmission congestion can quickly lead to localized distribution curtailment[18]. Integrating real-time grid frequency and power-flow telemetry with geospatial transit matrices and weather alerts creates an intricate data space that rural emergency managers cannot interpret manually during an unfolding crisis.

3. Why a Multi-Agent State Machine is Architecturally Mandatory

Managing emergency infrastructure requires reconciling electrical engineering models, geospatial transit mechanics, and public health epidemiology. A simple generative model or linear script cannot process these multi-system trade-offs safely:

Pipeline Node Identity	Node Classification	Operational Responsibility	Input Dependencies	Emitted State Channels
Grid Telemetry & Fragility Worker	Worker Node	Interfaces with the EIA v2 API via FastMCP tools to track hourly demand spikes, reserve margins, and corridor congestion[18]	Balancing authority ID, historical baselines	grid_stress_index, outage_risk_corridors
EMS Isochrone & Routing Worker	Worker Node	Calculates real-time travel isochrones over road networks, integrating weather-impaired routing barriers	Road networks, NOAA alert polygons	ems_isochrones, transit_deadzones
Vulnerable Population	Worker Node	Maps regional HHS emPOWER	Zip codes, emPOWER	high_risk_clusters, triage_prioritie

Worker		data and Census demographics against grid stress and response gaps	counts	s
Critical Infrastructure Evaluator Node	Evaluator-Optimizer Loop	Enforces physical power balance conservation, validates travel routing safety, and audits triage logic[3]	Worker outputs across all channels	verified_resilience_plan, optimization_retry

The Evaluator-Optimizer loop ensures that operational recommendations remain physically viable and medically safe[3]. In emergency response scenarios, inaccurate dispatch routes or faulty risk assessments can cost lives. The Evaluator Node verifies power calculations, ensuring that balancing authority demand and interchange calculations follow basic physical conservation laws:

$$\sum P_{\text{generation}} - \sum P_{\text{demand}} - P_{\text{losses}} = \Delta P_{\text{interchange}}$$

The Evaluator also checks that proposed emergency vehicle routes bypass active hazard polygons (such as flooded roads or damaged utility lines) and confirms that triage priorities align with HHS emPOWER life-safety criteria. If an agent recommends an impossible transit route or miscalculates grid capacity margins, the Evaluator rejects the plan and routes execution back for route adjustments[3].

State persistence is maintained using AsyncPostgresSaver, saving intermediate execution states within PostgreSQL[1]. This allows the system to maintain situational awareness across shifting weather patterns and multi-hour power outages without state corruption or session loss[1].

4. Admissions Impact & Faculty Interview Narrative

The candidate frames the project as a systemic intervention for neglected rural infrastructure. The narrative describes experiencing an extended rural power outage or observing rural EMS delays firsthand, motivating a deep investigation into how civic telemetry can be integrated to protect vulnerable populations. This narrative demonstrates technical ambition, systems-level problem solving, and a clear dedication to public service.

During an interview, a faculty member may inquire about how the architecture handles latency

and maintains consistency when live EIA grid telemetry and NOAA weather APIs drop out during severe storms. The candidate defends the resilience pattern: external API tools are wrapped in asynchronous circuit-breaker patterns. If the live EIA v2 endpoint drops during a storm, the system automatically falls back to an autoregressive moving-average projection derived from historical seasonal loads stored in the local PostgreSQL database[18].
The checkpointer records this degraded state in checkpoint_blobs, adjusting the confidence metrics passed to the Evaluator-Optimizer loop[1]. This guarantees that emergency managers receive defensible operational plans even under partial communication blackouts.

5. User Experience (React + FastAPI Integration)

A county emergency operations director or rural EMS dispatcher views the platform through an operational dashboard built with React and styled with Tailwind CSS. The user selects a county or balancing authority jurisdiction to monitor real-time infrastructure conditions.
FastAPI coordinates data ingestion and graph execution in the background, streaming updates to the dashboard via WebSockets:

- An interactive Geospatial Situational Map renders real-time EMS ambulance coverage isochrones, overlaying active NOAA severe weather warning polygons.
- A Regional Grid Stress Monitor displays real-time telemetry from the EIA v2 API, illustrating hourly load versus available capacity and highlighting vulnerable distribution areas[18].
- A High-Risk Population Triage Panel highlights geographic clusters where power-dependent medical populations overlap with extended EMS response dead-zones and imminent grid failure risks.
- An Automated Resource Allocation Engine generates prioritized mutual-aid dispatch recommendations and mobile generator deployment routes, providing emergency teams with clear, actionable operational guidance.

6. Feasibility Audit (4-Week Engineering Sprint)

The sprint MVP implements the pipeline for a dual-county rural region (such as Bastrop County, Texas or a rural section of California). The system ingests real-time balancing authority demand telemetry from the EIA v2 API (ERCOT or CAISO) alongside active NOAA weather feeds[18]. Using local road network data, it calculates 10-, 20-, and 30-minute EMS response zones, overlaying HHS emPOWER aggregate counts to output prioritized emergency shelter deployment maps.
The primary failure mode involves telemetry packet drops and rate-limiting across federal APIs during extreme weather crises[18]. The ingestion layer handles this through an asynchronous client that enforces strict 1,500ms timeouts and caches recent responses in PostgreSQL. If an external API fails, the pipeline degrades gracefully by using cached data and appending visual confidence warnings to the React interface.

Systems Architecture, Persistence Plumbing, and Technical Interview Defense

Across all five system blueprints, high-stakes public problems are resolved by coupling asynchronous Python backends with deterministic relational persistence and multi-agent cyclic verification.

Architectural Layer	Implementation Technology	Primary Functional Mandate	Defensive Failure Mode Strategy
Presentation Tier	React.js, Tailwind CSS, TanStack React Query	Visualizes real-time state steps, streaming reasoning, and verified legal/risk ledgers	Asynchronous polling with exponential backoff and cached state fallbacks
Ingestion Tier	Python 3.11+, FastAPI, Pydantic v2	Manages asynchronous endpoint lifecycles and enforces strict input validation schemas	Emits HTTP 422 validation errors before stochastic execution occurs
Orchestration Tier	LangGraph (StateGraph, Send, Conditional Edges)[3]	Directs task execution between orchestrators, workers, and evaluators[3]	Bounded recursion limits prevent infinite loops
Tool Protocol Tier	FastMCP (Model Context Protocol), REST APIs[4]	Decouples external data sources and local analytical engines into standardized tools[4]	Circuit-breaker timeouts fall back to local relational cache tables
Persistence Tier	PostgreSQL, AsyncPostgresSaver, psycopg_pool [cite: 1, 2, 20]	Maintains thread-level state across checkpoints, checkpoint_blobs, and checkpoint_writes [cite: 1, 20]	Database transactions ensure recovery without data loss after process restarts[20]

PostgreSQL Persistence Plumbing

Durable state persistence across all five systems is managed by langgraph-checkpoint-postgres[1]. In production environments, managing high-throughput

asynchronous agent threads requires careful connection pooling. Deploying AsyncPostgresSaver with psycopg_pool.AsyncConnectionPool prevents connection exhaustion and database deadlocks during parallel worker operations[2].
During FastAPI application startup, an async lifespan handler initializes an AsyncConnectionPool configured with dictionary-based row factories and autocommit modes[21]. The checkpointer's setup routine runs automatically, executing migrations that provision the core persistence tables: checkpoints, checkpoint_blobs, checkpoint_writes, and checkpoint_migrations[20].
The underlying PostgreSQL database schema maintains state across distinct operational tables:

- The checkpoints table records snapshots of the graph state at each superstep, indexed by thread identifier, checkpoint namespace, and monotonically increasing checkpoint IDs[1].
- The checkpoint_blobs table stores large, serialized channel payloads out-of-band, preserving query speed across the primary checkpoint index[20].
- The checkpoint_writes table captures intermediate node writes during graph execution, ensuring that partial work completed by parallel workers is preserved even if a downstream node encounters an exception[1].

State channels are structured using typed dictionaries. Scalar values (such as execution flags and loop iteration counters) overwrite cleanly across steps, while audit trails, violation records, and evidence arrays use append reducers (operator.add) to accumulate a complete, immutable record of the investigation[8].

Technical Panel Defense Strategy

During technical evaluations with admissions committees or competition judges, candidates can defend their architectural choices along three primary dimensions:

- **Separation of Stochastic Inference and Deterministic Verification:** The candidate emphasizes that language models are never permitted to perform numerical arithmetic, calculate legal deadlines, or make unconstrained decisions. Language models are restricted to unstructured entity extraction and narrative synthesis. All mathematical sums, spatial operations, and statutory boundary checks are executed through deterministic Python functions and parameterized SQL queries, enforced by a cyclic Evaluator-Optimizer loop[3].
- **Resilience Against State Drift and Infinite Loops:** The candidate explains how the cyclic graph prevents non-terminating loops. Every StateGraph incorporates an explicit recursion limiter and a bounded iteration counter. If an Evaluator-Optimizer loop fails to achieve invariant compliance within a specified threshold (typically three iterations), the graph gracefully exits to a fallback state, flagging the exact discrepancy in PostgreSQL for human review.
- **Auditability and State Reconstruction:** The candidate explains how AsyncPostgresSaver enables post-hoc forensic auditing[1]. Because every superstep writes an immutable checkpoint to PostgreSQL, developers, caseworkers, or judicial officers can trace the exact chain of reasoning, reconstruct prior graph states, and verify

that all outputs are grounded in authoritative federal open data[1].

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